The Exhibit.

The Gold Car Heating & Lighting Company, New York, has been demonstrating, at its booth, the new Ideal pressure reducing valve for car heating It is claimed for the regulator that when once set it will not vary.

The new combination body and truck bolster car which has been shown among the track exhibits by the Mid-Western Car Supply Company, Chicago, will be routed to Chicago via the Reading and Lackawanna roads, leaving Atlantic City Thursday. The car will be tested at Chicago.

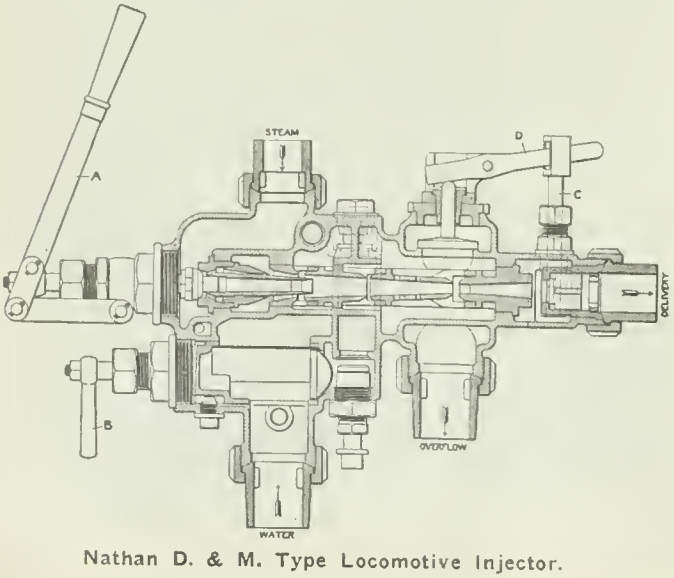
The Acme Supply Company, Chicago, at spaces 572 and 573, is making the first exhibit of Chanarch car floor plates. These plates are used with any of the composition floorings usually found in up-to-date passenger equipment. Chanarch automatically anchors the composition and weighs less and requires less composition than other floor plates, thus effecting a saving not only in cost, but operating efficiency. The Acme company will enlarge its plant on West Lake street and largely increase its output.

INJECTOR WITH AUTOMATIC OVERFLOW VALVE.

Since the invention of the injector by M. Giffard there has been a constant endeavor to improve its design and construction in accordance with the tendency toward increased boiler capacities and higher steam pressures. The first non-lifting injector manufactured by the Nathan Manufacturing Company, New York, consisting merely of a body and a set of nozzles, was followed by lifting injectors of the fixed nozzle type, the restarting type and of the non-lifting type, in which all operating parts necessary to start, regulate and stop it were contained in the injector itself.

All these types have their advantages, but also have weaknesses, one of which is the tendency to spill water at the overflow when the temperature of the feed water becomes higher than is suitable to the characteristics of the injector. This tendency has been overcome, in most cases, by what are called closed overflow injectors, in which the overflow is closed by hand-operated devices. This arrangement is unsatisfactory, and under certain circumstances may even be dangerous, because, if the jet is broken, the steam is blown back in the water tank, and, if it is not noticed, the water in the tank will soon become too hot to be handled.

All difficulties and uncertainties of this kind have been overcome in the D. and M. type injector, illustrated herewith, which represents an entirely new departure in injector construction shortly to be introduced by the Nathan Manufacturing Company.



Nathan D. & M. Type Locomotive Injector.

This new type is, in general, similar to the restarting type of Simplex instrument, but includes an automatic closing and opening overflow valve. Letter *A* designates the ordinary starting lever, operating in the usual way, and *B* indicates the water valve by which the admission of water is regulated according to requirements. The injector is provided with the usual overflow valves and an auxiliary inlet valve. In the delivery chamber, moving in a suitably packed stuffing box, a small piston, *C*, which actuates a fulcrumed lever, *D*, that at the proper time acts on the overflow valve.

When the injector is started, the pressure in the delivery chamber slightly raises the piston, *C*, and the lever *D* bears down on the overflow valve and keeps it closed. Should the injector break from some cause, such as shortage of water, the pressure under the piston ceases and the overflow valve will open automatically. The injector then restarts automatically as soon as the water supply is restored. Closing the overflow is automatically repeated after the injector is again started.

This injector thus combines within itself the advantages of both the closed overflow and the open overflow injector, and entirely eliminates the disadvantages of the forcibly closed overflow. In view of the overflow being closed automatically, the injector is enabled to work feed water as hot as any type of injector with a hand operated overflow valve.

CUTTING COMPOUND.

An oil that is a distinct departure from the ordinary cutting compounds, is being introduced by Flint & Chester, Inc., New York. Contrary to the general characteristics of oil it thoroughly mixes with water and forms a solution that has all the advantages of water and at the same time the lubricating qualities of oil. It is known that a globule of water will resist a greater pressure than an oil globule, the difficulty in employing water as a lubricant being its lack of viscosity, or ability of globules to cling together.

As a mixture of from 10 to 20 parts of water to one of oil, depending on the character of work, forms the cutting compound, the cost of the mixture per gallon is small and it has the added advantage of cheap handling and storage, for the freight charges on and space occupied by one barrel of the oil cover a final mixture from 11 to 21 barrels. This oil is also valuable for hydraulic machinery. By using from 50 to 75 parts of water with one of oil it forms a preservative for the machinery as well as a lubricant.

The mixture does not give off the usual disagreeable odor when heated; it prevents rust, permits of heavier cuts being taken, and has healing instead of the poisonous qualities.

THE WESTINGHOUSE EXHIBIT.

Contrary to the practice of previous years, the several Westinghouse companies have had no working exhibit, their booth in the entrance hall being tastefully furnished as a place where members and guests of the convention could meet and talk. However, Westinghouse motors with control are being used in connection with the machinery exhibits of the following concerns: American Tool Works, 3 h.p. "CCL" motor driving a 2 ft. geared radial drill; a 20 h.p. "CCL" motor driving a 6 ft. radial drill. Baker Brothers, an "SK" adjustable speed DC motor on a 1 in. x 6 in. automatic drilling machine. Bethlehem Steel Company, two "CCL" motors operating a stress testing machine and a staybolt vibrating machine. Bullard Machine Tool Company, a 20 h.p. "CCL" motor on a 42 in. vertical turret lathe. H. H. Hewitt, two type "R" direct current motors on truck. Manning, Maxwell & Moore, Inc., an "SK" motor with drum type controller mounted on a Reed lathe, and an "SK" motor driving a National multiple spindle topping machine. Special attention is called to the automatic starter with this motor. Clement Street, two Westinghouse air brake motor-driven air compressors. Vixen Tool Company, a "CCL" motor driv-

Proceedings.

Vice-President Crawford called the meeting to order at 9.55 a. m.

FLANGE LUBRICATION.

The committee reporting last year on the subject of flange lubrication has been continued to make still further report to the 1912 convention and the original committee has been increased from three to five members.

Since closing the 1911 report advice has been received of flange lubricators having been applied by over thirty roads on which lubricators had not been used previously. In most cases reported, the number of locomotives equipped on each road has been small. In one instance, however, a road operating over 1,100 locomotives has applied flange lubricators to the entire number.

On a number of other roads on which flange lubricators were used at the time of preparing the last report, the application



M. H. HAIG,
Chairman, Committe on Flange Lubrication.

has since been extended. The reports of thirty-five of these roads show an average increase of thirty lubricators each, the number per road varying from one to over two hundred.

An affirmative answer to the question "Does your experience confirm the report on flange lubrication presented at the last convention?" was received from twenty-three of the twenty-nine roads replying in detail to the committee's circular. Five of these roads explained that failure to answer was due to lack of sufficient experience.

The types of lubricators applied during the past year indicate the most general adoption of crude oil as a lubricant. Some reports state that hard grease has been rejected as unsatisfactory, due to the fact that the grease catches and holds sand. The abrasive action resulting has increased instead of decreased the flange wear. Graphite as a flange lubricant has been reported as successful by some roads but others have abandoned it for crude oil. Crude oil with a heavy asphaltum base is spoken of with the greatest favor by roads using flange lubricators. It has been successfully applied by both hydrostatic and gravity-feed devices. Exhaust steam from air pumps as well as water jets are still being used to a limited extent by some railways and beneficial results are reported. An undesirable feature of the water jet is the fact that it must be abandoned during the winter weather.

A type of lubricator especially adapted to electric locomotive service has been developed on the Grand Trunk by W. D. Hall, superintendent of power plant and electrical equipment. This lubricator has been designed to meet the requirements of service through the St Clair tunnel. Fig 1 is a sketch illustrating the principle by which this lubricator is connected and operated. Oil is contained in an air-tight receptacle of one quart capacity. The oil is led to the wheel flanges by pipes and is sprayed upon the flanges by jets of air. The air is supplied from the main reservoir by a ¼-in. pipe connected to the oil receptacle above the surface of the oil. This provides a pressure to force the oil from the receptacle. A branch of this pipe is connected to the oil delivery pipe leading to the flanges. The oil is thus forced through the pipes and sprayed upon the flanges.

Used with electrical equipment, the air is controlled by an electric push-button. Valves admitting air above the surface of the oil, as well as to the delivery pipes are operated by electromagnets. The push-button is located in easy access of the engineer and oil is sprayed upon the wheel flanges almost

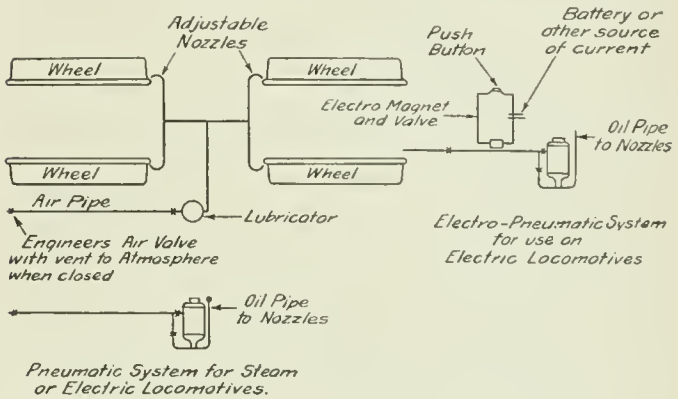


Fig. 1—Pneumatic Flange Lubricator for Electric Locomotives; Grand Trunk.

immediately upon the admission of air pressure. With this arrangement flange lubricant is applied only when needed and when running on straight track the lubricator is not continued in operation.

This apparatus has been in successful operation since July 10, 1910, and has operated under extremely severe conditions. Six electric locomotives have been equipped and handle 1,000-ton trains up and down 2 per cent. grades. Owing to the many curves encountered and also because of the rather slow center of gravity of the locomotives, flange wear has been quite heavy. Since applying this lubricator the conditions have been improved to such an extent that 50,000 miles and over are made between turnings and it is seldom necessary to remove wheels for tire turning before the armature is removed for commutator dressing, and these two operations may, therefore, be performed at the same time.

Before the application of the lubricator, mileage made between tire turnings was from 12,000 to 25,000. The lubricant used is engine oil, most of which is taken from bearing cellars of the railway armatures when the oil in them is being changed. The oil is filtered before being applied to the flange lubricators. No difficulty has been experienced with the delivery of this oil in cold weather. It is sometimes thinned to suit the weather conditions by the addition of a little coal oil. Care has been exercised in keeping the oil free from anything that might obstruct

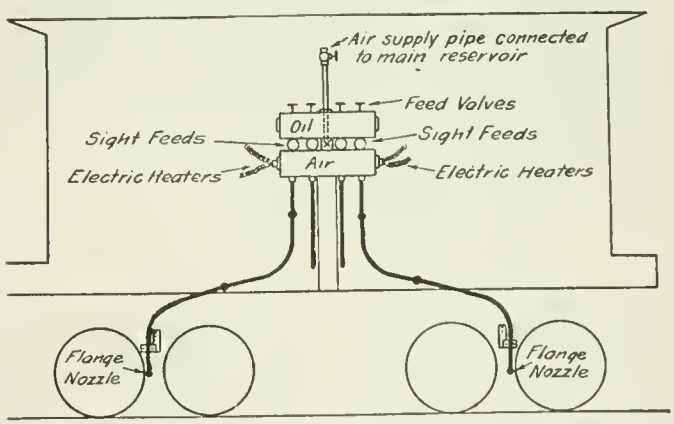


Fig. 2—Electric 4 Pneumatic Flange Lubricator for Electric Locomotives.

the nozzle and very satisfactory results have been obtained under varying and severe weather conditions.

Fig. 2 is a diagram showing the application and principle of operation of an air-operated flange lubricator for electrical equipment, manufactured by the makers of an hydrostatic flange lu-

bricator for steam locomotives. The lubricator is located in the cab as with the hydrostatic device, and sight feeds indicate the rate of formation of the drops of oil. An electrical heater is included in the equipment and the temperature of air and oil is raised by passing through this heater. The air supply is from the main reservoir. Oil is led to wheel flanges by pipes as with the hydrostatic lubricator and is sprayed upon the flanges by air instead of steam. This lubricator is designated by the manufacturers as the "Electric 4." It has been placed in service on electric locomotives of the Baltimore & Ohio and the Pennsylvania.

The Collins flange lubricator illustrated and described in the former report has since been improved. The improved device is similar in construction to that already described. The feeding device of the original type was placed under the body of the lubricator. The lubricating block of graphite composition was pressed against the flange by a pawl engaging in notches in the under side of the block. With the improved type a coil spring and plunger acting directly behind the block of lubricant presses it against the flange, the feed being continuous without adjustment until refilling is necessary. The 1911 report has given the impression that this lubricator was designed to use hard grease. It should be stated that the lubricant is graphite.

Among the simple devices operated by some of the railway companies is one which can be built up of pipe fittings and has been used to apply either hard grease or graphite. The device consists of a short section of 1¼-in. pipe capped at one end and containing a piston actuated by a spiral spring. A small rod threaded into the piston extends through the cap, not only assisting to guide the piston but also indicating the amount of grease in the lubricator. This device is clamped to the frame of the locomotive in front of the drivers and feeds the lubricant directly against the flange. When using ordinary rod grease the lubricant is contained in a section of scrap rubber hose and is deposited as the hose is worn away.

The Pittsburgh & Lake Erie has applied this lubricator extensively, using a mixture of paraffin and graphite as the lubricant. This is contained in a wooden block turned up in a lathe, the cavity for the grease being bored out 1 inch in diameter. The practice of this road has been to apply a lubricator to each wheel which has shown wear or cutting. After the flange has been worn smooth it has not been considered necessary to operate the lubricator until further signs of cutting develop. Some difficulty has been reported as to keeping this device in proper adjustment at all times.

The result of a test of flange lubrication on a number of twelve-wheel locomotives of the Norfolk & Western is contained in the 1911 report. This test led to the conclusion that with track conditions as existing on this line, flange wear was too slight on freight engines to warrant the practice of lubrication. Five Pacific type passenger engines have since been equipped with lubricators for test purposes. The equipment includes three types of lubricators—hydrostatic, gravity and graphite. The comparative mileage between tire removals before and after the application of the lubricators is shown in Table 1. It will be seen that the average service per front tire removed from these engines before the lubricators were applied was 16,518 miles, while the average for the same engines since the lubricators were installed has been 39,909. The former figures are taken from the record of 25 tires removed, the latter from the record of 10 tires removed. The local authorities providing the data suggest that this result may have been affected in some measure by a lateral spring arrangement applied to the trailer trucks to prevent nosing. They state, however, that they consider the improved condition to be due to lubrication and not to the spring.

TABLE 1.

MILEAGE SERVICE OF FRONT TIRES REMOVED FROM PACIFIC TYPE ENGINES TO FEBRUARY 1, 1912.

NORFOLK & WESTERN.

After Equipped with Oilers. Mileage.			Before Equipped with Oilers. Mileage.		
Eng.	Right Side.	Left Side.	Eng.	Right Side.	Left Side.
564	63,141	63,141	564	26,702	26,702
Average miles per tire, 63,141. Tires removed, 2.			Average miles per tire, 26,702. Tires removed, 2.		
569	27,889	27,889	569	13,175	13,175
569	39,582	39,582			
Average miles per tire, 33,735. Tires removed, 4.			Average miles per tire, 13,175. Tires removed, 2.		
571	26,728		571	17,204	17,204
571		43,512*	571	6,845	6,845
			571	12,096	12,096
			571	16,784	
Average miles per tire, 35,120. Tires removed, 2.			Average miles per tire, 12,725. Tires removed, 7.		
572	No tire renewals since oilers applied.		572	15,846	15,846
			572	24,553	24,553
573	33,816	33,816	572	24,297	24,297
			572	14,640	14,640
			Average miles per tire, 19,834. Tires removed, 8.		
			573	17,972	17,972
			573	15,618	15,618
Average miles per tire, 33,816. Tires removed, 2.			573	9,139	9,139
			Average miles per tire, 14,243. Tires removed, 6.		
			Average miles per tire, all engines, 16,518.		

Average miles per tire, all engines, 39,909.

*Includes four months' mileage before oiler was applied.

Fig. 3 shows the contour and mileage records of a decapod locomotive on the Buffalo, Rochester & Pittsburgh. Previous to the application of the lubricator this engine had made 31,529 miles with the flanges worn as shown by the dot and dash con-

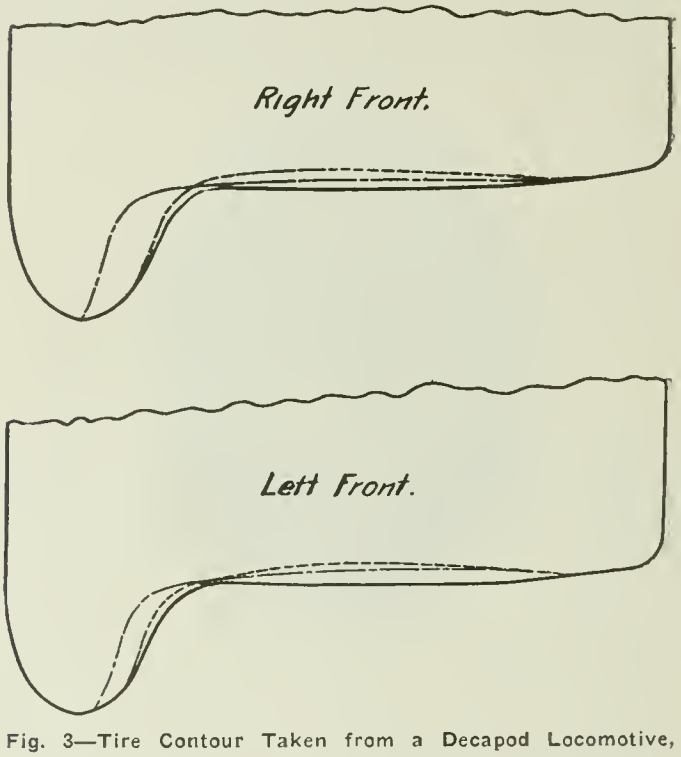


Fig. 3—Tire Contour Taken from a Decapod Locomotive, B. R. & P.

Full lines, standard tread and flange as received new from builders, Sept., 1909, also represents new tires applied July 28, 1910, at which date the Chicago Flange Oiler was applied.
Dot and dash lines, tread and flange July 28, 1910, after making 31,529 miles (previous to applying flange oiler).
Dotted line, tread and flange March 9, 1911, after making 21,102 miles (Engine equipped with Chicago Flange Oiler).
Contours taken from Plaster Paris casts.

tour lines. After making 21,102 miles with a hydrostatic oiler in service the condition of tires was as shown by the dotted contour lines.

The Hocking Valley Railway reports that the service of switch engines operating upon industrial tracks with much sharp curvature has been increased from 4 months to 12 months. On consolidation engines subject to excessive wear of front and back flanges the service of tires has been increased from one year to two years. At the end of this period the flanges are still in good condition.

In Table 2 is shown the estimated saving in cost of fires effected by the use of flange lubrication on the St. Louis & San Francisco. This is based upon experience with consolidation engines equipped with hydrostatic oilers using rasp altum oil.

TABLE 2.

SAVING IN TIRES ON CONSOLIDATION LOCOMOTIVES EQUIPPED WITH FLANGE LUBRICATORS.

ST. LOUIS & SAN FRANCISCO RAILROAD.

Value of new 56-in. tire, 3½ in. thick	$30.80
Less scrap metal in tire	2.45
Net value of wearing metal	$28.35
Net value per sixteenth, figuring on 28 sixteenth, wearing metal	1.01

as possible, as developed at that time. It therefore now feels that its further duty lies in the recording of the developments in flange lubrication during the last year, the increase or decrease in flange lubrication, as well as reporting the further actual experience of the various railroads, thereby either confirming or refuting the previous report. Since last reporting, there has been some development in the line of flange lubrication for electric locomotives. Two lubricators for this class of work are described in this report. Some impression has been current that the oil used in flange lubricators which use oil as a lubricant, is provided also exclusively from a certain source. It should be pointed out for the benefit of all members that the oil is available in the most convenient market. It has been found, though, from experiment that the Kern River oil from the Bakersfield district of California, is the most satisfactory for the reason that it adheres to the flanges and forms a sort of paste that not only spreads over the flange, but is wiped off against the rail where it forms a coating. In the hydrostatic lubricator, the oil being a heavy body, will not separate when it comes in contact with steam.

Very often information of unusual interest is contained in the letters that the various members forward to committees, and while this information cannot always be given in full in the reports, I feel that a few quotations would be of interest. For instance, with regard to the delivery of lubricant to the rail. A member states that from personal observation in examining the ball of the rail after train has passed, and when the flange lubricator is properly operated on the engine, it is possible to see a slight indication of oil on the rail which he states proves that the ball of the rail was lubricated to a certain extent; a condition which should reduce wear of rail as well as wear of flanges.

Another member in answering the various questions advises that flange lubrication prolongs the life of the rail by one-third, and goes on to state that anything that will save the flanges will save the rail. With regard to comparative wear of flanges between turnings with and without flange lubricators, I refer to a statement made by a member concerning both passenger and freight equipment. He states that engines working exclusively in mountain service require tire change every 8,000 to 10,000 miles when operated without flange lubricators. He then refers to a ten-wheel passenger engine making 37,186 miles before turning tires, and to a Mikado freight engine which made a mileage of 21,152 miles up to the time of a wreck, and examination of the tires and flanges showed them to be good for, at least, 2 months more.

Another member advises that since the application of flange lubricators to a number of ten-wheel passenger locomotives, it has been possible to make from 65,000 to 85,000 miles per engine. With the same class of engine without flange oiler it has been necessary to turn tires on account of flange wear, after making from 25,000 to 30,000 miles. A number of replies indicate particularly satisfactory results with switch engines. Engines of this class work a great deal over industrial tracks in which the curves are sharp and frequent, and various reports indicate that the life of tires is much longer with flange lubricators than without. For instance, one member advises that the service of switch engines is increased from 4 to 12 months.

The committee was particularly anxious to see that all lubricators which had attained any success were described in this report. A short time before closing the report, we received word from one member of the association that he was using a Detroit lubricator. This lubricator is a gravity feed device. The tank is supported under the running board on one side of the engine and a pipe leads to one flange on each side of the engine. The valve which controls the flow of the oil is operated by a ball which receives its motion from the movement of the locomotive.

DISCUSSION.

C. A. Seley (C. R. I. & P.): I cannot understand what the third recommendation means.

Mr. Haig: The lubricator is naturally located on the locomotive. Usually it is located on the driving wheel, though occasionally it has been placed on the engine truck wheel. Instead of stating that the feed of the lubricant decreases as the distance from the locomotive increases, it might have been plainer had we stated that it decreases as the distance from the point of application increases; the lubricator naturally has its greatest effect on the wheel to which it is applied, inasmuch as it spreads the lubricant very fully over that flange. The flange conveys the lubricant to the rail, and as the train passes over that point each wheel picks up a little of the lubricant, so that by the time the last wheel of the train passes that point, it receives less benefit from the lubricant than the first wheel. The object is to spread a paste over the side of the rail, if possible, in order that not only the rail but all of the wheel flanges will receive some benefit.

F. F. Gaines (C. of Ga.): We have tried out pretty nearly everything in the way of flange oilers except the last one referred to by Mr. Haig, and we have boiled the matter down to the hydrostatic form of lubricator as being the only satisfactory lubricator. On one of our divisions we have had quite some considerable trouble, with our Pacific type engines, from the fact that the flanges would run sharp in the neighborhood of about 30,000 miles, practically, whereas after the application of the lubricator we find we can get about 60,000 miles. After equipping these engines with the lubricator, we find we can get 60,000 miles out of them, and figuring on the tire turning and loss of metal involved, and the labor and material, at $250 for each turning, the improved mileage means a very considerable economy, amounting to over $500 between each tire turning on these locomotives.

On another division we equipped the entire system with flange oilers, and our figures, while not showing the saving that some of the roads reporting show, averaged up to $50 an engine per year on this division, although our figures are not yet as complete as they might be, due to the fact that we have only recently got the whole division equipped. We expect to have more definite figures in the future. As an illustration, I will say that the value of tire loss on 8 tires, in the case of engines equipped with the flange oiler, was $74.88, whereas the value of tire loss on 8 tires, in case of engines not equipped with the flange oiler, the loss of material involved is $132.48.

H. C. VanBuskirk (C. & S.): Our road has had considerable experience with flange lubricators of different types, and as Mr. Gaines has just stated, the hydrostatic type of lubricator is the only one, among those we have used, which gives satisfactory results. Our experience has been principally on the narrow gage lines of the Colorado & Southern, where there are long curvatures and high grade lines. There is one question that has been asked, and that is whether there is any effect on the flange lubricator in the case of trains descending on a heavy grade. I can say from my personal observation that there is not. We have tried that with the hydrostatic type of lubricator, by putting sufficient oil on the flange, which will communicate it to the rail, and there is no effect on the braking power when descending on a 4 per cent. grade. There is no noticeable effect. I have tried a lubricator which uses the solid lubricant, and it will pick up sufficient foreign matter so that it will glaze over the surface and give no result on the flange. We have 30 deg. curves, and lots of them, on 4 per cent. grades, and the locomotives in connection with which we are now using the hydrostatic lubricator will run from one shopping to the next, whereas with the lubricator we formerly used, we could never get over 12,000 to 14,000 miles out of our locomotives. At the present time we are running in the neighborhood of 50,000 miles, and never have to take the locomotives out of service on account of flange wear. The gravity type of lubricator that we have been trying will go about 5,000 to 12,000 ft. elevation on one division in warm weather, but by the time we get at the top of a mountain the lubricator will not feed in the same manner, whereas the hydrostatic is under control at all times and will deliver the oil just as it is wanted.

R. D. Smith (B. & A.): The report of the committee at the last year's convention was such as to prompt us in trying these flange lubricators on some of our Pacific type engines; and the results that we have obtained have been very good. I have, however heard a criticism of it since being here, with which we have not had any trouble, and that is, there is a slippage of the engine when starting a train due to the spurting of the oil while the engine is standing. I mention this from a hydrostatic lubricator point. The results that we have obtained have been so good as to warrant us in extending their use.

Angus Sinclair: I think I was one of the pioneers in experimenting with flange lubrication. About 1880 I was running a locomotive on a small road. They had only one engine, and the road was very crooked and it ended with a curve of about 15 deg. or so, and I hadn't been running the engine but a short time when I found flange cutting going on. I was aware that if the engine failed, my job would fail, for they would not ever be able to buy another, and consequently I was very much interested in preventing this flange wear. My first natural impulse was to try a swab that we used for piston rods, but that proved unsuccessful as it also lubricated the tread of the wheel. I finally fell upon a scheme of a piece of graphite and grease mixed, and I was fairly successful with that. I was able to prevent the flanges from cutting, and not altogether without lubricating the tread, but still not to any great extent, and I saved my job through the lubricating device, and I was rather proud of it in those days.

S. I. Bean (Santa Fe Lines): We have in my territory about 2,500 miles of road. The longest tangent is 30 miles. We have increased our mileage from 60,000 miles in the passenger

another member. Sling stays are used to the extent of 2, 3 and 4 rows back from the tube sheet on b iler with combustion chamber.

Flexible Staybolts.—Of the members who have replied, all except one member use flexible staybolts. Most of the members apply these in the breakage zone. Five members are making full installation of flexible staybolts on boilers with wide fireboxes, and one member uses them in the throat sheet, only. One member has the following report to make:

"In February, 1907, engine received a new firebox with a full installation of flexible staybolts, with the exception of 4 bolts under the auxiliary dome and 4 under the steam turret; also 8 bolts which go on top at the back head. February, 1908, the engine received general repairs and the firebox was in good condition. All caps were removed from the flexible staybolts and they were found to be o. k. In April, 1909, it was found necessary to patch the top of the back tube sheet on account of the sheet cracking from the hole around the flange; this was done in the engine house. In October, 1909, the engine was given general repairs, and by this time the top of the back tube sheet had given out in two more places, but as the sheet was in good condition otherwise it was patched. Three of the mud-ring corners also had to have small patches applied. After rattling the scale from the firebox sheets with a large hammer, it was found that 40 of the flexible sleeves had cracked outside of the sheet, and on being taken out they were found to be crystallized, but no broken bolts were found. In 1911, the engine was again in the shop for general repairs and had all the caps removed from the flexible staybolts and the bolts found to be o. k. A patch was applied on the top flange of the flue sheet, but the half side sheet seam in the firebox was in good condition. When the engine is shopped again it will be necessary to renew the back flue sheet on account of its being patched. This engine has made 319,567 miles to November, 1911, and has lost no time due to broken staybolts."

One member advises that they have equipped some fireboxes with a full installation of flexible staybolts, but can see no advantage in doing this. In their opinion the flexible staybolts are only advantageous when applied in the breakage zone, and

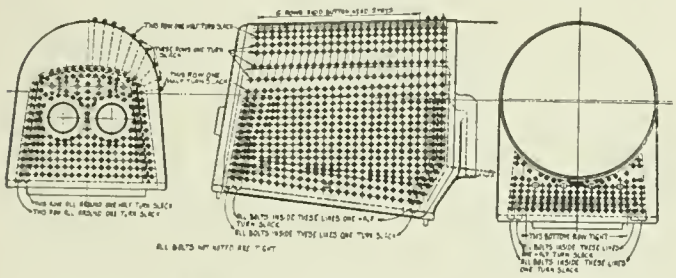


Fig. 13—Rivet Spacing Showing Location of Flexible Stay bolts.

they have discontinued making full installations and are only applying them in the breakage zone. However, a full installation of flexible staybolts in the throat sheet is being used by quite a few members with good results.

Three members advise that they allow a certain amount of slack when applying flexible staybolts. Fig. 13 shows the amount of slack and the location followed out by one of the members. It is the brief that an application of this kind will decrease the number of broken staybolts and allow for a greater expansion and contraction, which prevents the cracking of firebox sheets and increases the life of firebox correspondingly.

The committee believes that this is a subject worthy of further consideration and would suggest that some more of the members carry on a further investigation and make a report at some future date to this Association.

Tube.—It is the consensus of opinion of those reporting that 2-in. tubes are preferred, until the length of tube exceeds 16 ft., when 2¼-in. tubes are then preferred.

Taking into consideration that a 2-in. tube will give a greater amount of tube heating surface, and the cost of maintenance is less, and less damage is done to the tube sheets in working over the flues, the committee suggests that further consideration be given to the use of 2-in. tubes in lengths greater than 16 ft.

Thickness of Front Tube Sheet.—One-half inch and ⅝ in. are used by the majority of the members who replied to the circular. One member uses 9/16 in. and another ¾ in.

Thickness of Back Tube Sheet.—One-half inch tube sheets are being used by 16 of the members. Two members use 9/16 in. and one member uses ⅝ in. All but three report having more or less trouble with flue sheets working up on account of frequently working on the flues, causing the flue sheets to crack across the top of the flange of the sheet. One member reports that this trouble occurs only on the radial type fireboxes. The crown bar and Belpaire type never crack in this way. The top flanges get bent down on the two latter types, the same as in the radial type, but no cracking takes place. They have used eye bolts in crown sheet and roof sheet and sling braces between; also have a number of boilers equipped with flexible expansion stays in the first four rows across the front end of crown sheet, but there does not seem to be any method of bracing yet devised that will overcome this trouble. He suggested that the top of the flue sheet on radial type fireboxes be made straight across top for 24 in. each side of center and make front end of crown sheet conform to this shape.

Width of Bridges.—The width of bridge in the front tube sheet varies from ½ in. to 15/16 in. Three-fourths inch is used by the majority of the members. Width of bridge in the back tube sheet varies from 11/16 in. to 15/16 in. Three-fourths inch is used by the majority of the members.

Flue Setting.—Two members advise that they have tried out soft iron ferrules with very little success. One member advises that they had a limited experience trying to set flues in the back flue sheet without any coppers, but found that it did not work satisfactorily, and that they are now experimenting with a combined copper and iron ferrule, but have not had these under a test long enough to make any report. One member advises that they use soft iron shims on the front flues to avoid excessive expanding.

Setting of Flues in the Front Tube Sheet.—The consensus of opinion of the roads reporting is that tubes should be rolled in the front tube sheet without any ferrules, and 10 per cent. of the flues, equally distributed, should be beaded with a beading tool.

Tube Setting in Back Sheet.—Fig. 14 show the practice fol-

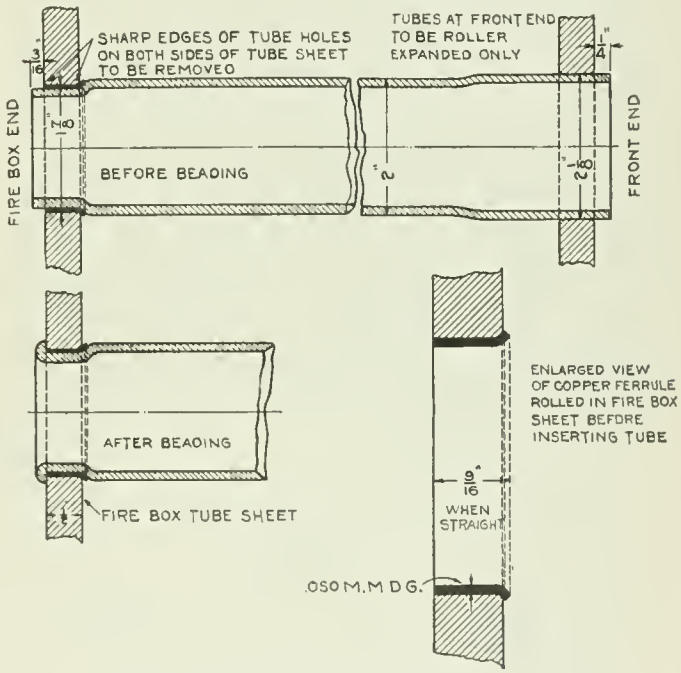


Fig. 14—A Method of Setting. Flues.

lowed out by some of the members in setting the tubes, which are giving good results. Fig. 15 shows the application, maintenance and removal of flues in locomotive boilers as used by three members. Tube setting of this kind gives a large bridge, good circulation of water at the back tube sheet and at the same time the inside of the flue is reduced at the end, so that any cinders which will pass through the swedged end will not clog the flues. The instructions in connection with Fig. 15 are as follows:

Holes in the new firebox flue sheet shall be 1⅞ in. diameter. Holes in the new smokebox flue sheet shall be 2 1/32 in. diameter. Holes in an old flue sheet more than 1/32 in. out of round must be reamed (reamer Fig 1). Inside and outside edges of the flue holes in both sheets must be slightly rounded to remove any sharp edges. The following is the method used:

Flue-sheet Holes (Fig. 1)—Care should be taken that holes in flue sheet are true, smooth and free from burrs and sharp edges. It is desirable that the flue hole have a fillet especially on the water side of about 1/32-in. radius. A sharp edge around the hole often cuts the ferrule in two, even cutting into the flue. The diameter of the flue hole should be the same as outside diameter of flue plus 1/32 in.

Copper Ferrule (Fig. 2)—For New York this should not be far different from No. 16 B. W. G. In old flue sheets when the flue holes are large it is desirable to use enough heavier ferrules to bring the internal diameter

of ferrule when expanded into flue sheet to 1/8 in. less than flue diameter. The ferrule should be set into flue hole flush with fireside and expanded into place with Prosser expander, Fig. 3.

Swaging Flue (Fig. 4)—Great care should be taken to give the swage the right length and to have it terminate in as abrupt a shoulder as possible. The flue, after it is swaged, should be annealed and the scale removed from the portion entering the sheet. Grinding this scale off by machine is desirable, but in the absence of a grinder the removal of the scale with a file will answer. The flue when driven into the sheet (Fig. 5) should extend 3/16 in. through the sheet on the fire side.

Rolling (Fig. 6)—The front end of the flues having been shimmed, when necessary, both the back and front ends are rolled with the dudgeon or other suitable roller, with proper appliances they may be safely and economically rolled by air; the rolling of the back ends must be done by an experienced boilermaker, and the motor must not be larger than a Thor No. 22 or its equivalent.

Beading (Fig. 7)—This may now be done with a beading tool in an air hammer. Do as little hammering with the beading tool as possible, and do not work bead to a feather edge, but leave as much metal as possible for future caulking.

Expanding (Fig. 8)—Particular stress should be placed upon this, the most important step in the flue-setting process. Care should be taken that the flue shoulder when heavily expanded bears snugly and firmly against the flue sheet. Hammer not to exceed 4 lbs. after prossering the front ends, then bead a sufficient number for safety which finishes the flue setting. Back ends of flues or safe ends when used in back flue sheet must be cut off square in pipe machine and burr removed by reamer. Wheel-flue cutters must not be used on account of leaving a heavy burr which splits when beading over.

NOTE.—Air motor and roller are only to be used on initial setting. In reworking flues, the Prosser expander only is to be used, and rollers must not be used, as they force the bead away from the sheet.

Copper ferrules 1¾ in. inside diameter, ½ in. long, .095 in. in thickness shall be used in firebox flue sheet only. Ferrules shall be secured in place with straight expander, Fig. 2, taking care that shoulder of expander is tight against flue sheet, which shall bring edge of ferrule 1/32 in. from fire side of sheet.

Flue safe ends should be 5 in. in length, except that new iron flues shall have safe ends 4 in. long, and when these flues are first removed the 4-in. ends will be cut off completely and 6-in. ends applied, after which 5-in. ends should be used. Flues shall be swaged as per Fig. 4, to 1 11/16 in. outside diameter by 3 in. long, including taper at firebox end, for new work; for old work, just enough to enter the copper ferrule. All scale must be removed from swaged ends of flues before application. Smokebox end of flues shall not be more than 1/32 in. less in diameter than holes in the smokebox flue sheet. If necessary, liners may be used.

Flues shall be placed in back flue sheet with a bar, Fig. 5, and project ¼ in. through the sheet. Gage, Fig. 6, shall be used in checking location of flues. Flues shall then be fastened at firebox end with straight sectional expander, Fig. 7. While flue is being fastened it may be held in place at front end, using bar, Fig. 5. Straight sectional expander shall be checked with master gage, Fig. 8. Before using sectional expander, and in order to obtain exactly the proper length of flue for beading, flues may be rolled lightly to fasten them in sheet with roller expander, Fig. 15. After all flues are fastened they shall be flared at firebox end, using flaring tool, Fig. 9, and long stroke riveting hammer.

Flues at firebox end shall be expanded with sectional expander, Fig. 10, and long stroke riveting hammer. Roller expander must not be used. Pin shall be driven into expander until flue is solid against flue sheet. This must be done three times, expander to be slightly turned before each operation. The expanding shall be performed as follows: First, two vertical rows, from center to top of sheet. Second, same rows from center to bottom. Third, the two horizontal center rows from center to right. Fourth, same rows from center to left. Fifth, all remaining flues. All flues must be carefully inspected after expanding to assure that recess in each flue is the full depth to recess in expander and even all around the flue. Sectional expander, Fig. 10, to be checked with master gage, Fig. 11.

Flues in firebox flue sheet shall be beaded with standard beading tool, Fig. 12, and short stroke riveting hammer. Care must be taken so that nothing enters between head and flue sheet. Beading tool must conform accurately to master gage, Fig. 13, at all times. Smokebox end of flues shall be tightened with flaring tool, Fig. 14, before rolling takes place. All flues in smokebox end shall be rolled with roller expander, Fig. 15.

FLUE MAINTENANCE.

The firebox end of flues in service must be expanded at regular intervals with sectional expander, Fig. 10. This work shall be done when boiler is empty and all flues thoroughly cleaned out. Flue leaks in the firebox must be stopped with the sectional expander and not with roller expander nor beading tool. If heads are slightly away from flue sheet, standard beading tool shall be used to bring heads tight to sheet. Sectional expanders and beading tools must be kept standard by frequently comparing with standard gages. Any beading tool not conforming to gage must be sent to principal shop for repairs; these tools must not be repaired at any other shop. All shops and engine houses must be equipped with standard gages. All gages will be made where master gages are kept. To remove flues, use flue cutter, Fig. 16, for the front end, cutting off flues as close to the front flue sheet as possible. Back ends should be split as little as possible, so that it will not be necessary to use safe ends longer than 5 in.

Setting of Superheater Tubes.—Front Tube Sheet: Only eight members have had any experience with setting large superheater tubes. The practice followed out mostly is that

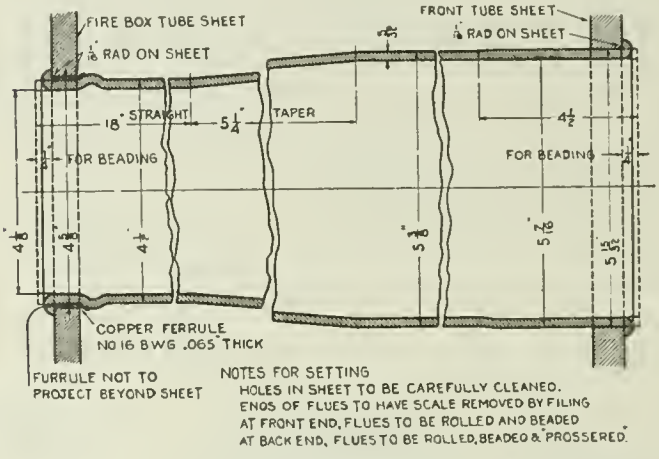


Fig. 16—Setting of Superheater Flues.

the tube is rolled and beaded as shown in Fig. 16, and this type of setting is suggested by the committee for further consideration.

Back Tube Sheet: Eight members advise that they use copper ferrules; one member who omitted copper ferrules is having good success with this method. The consensus of opinion of the roads reporting is that the type of setting shown in Fig.

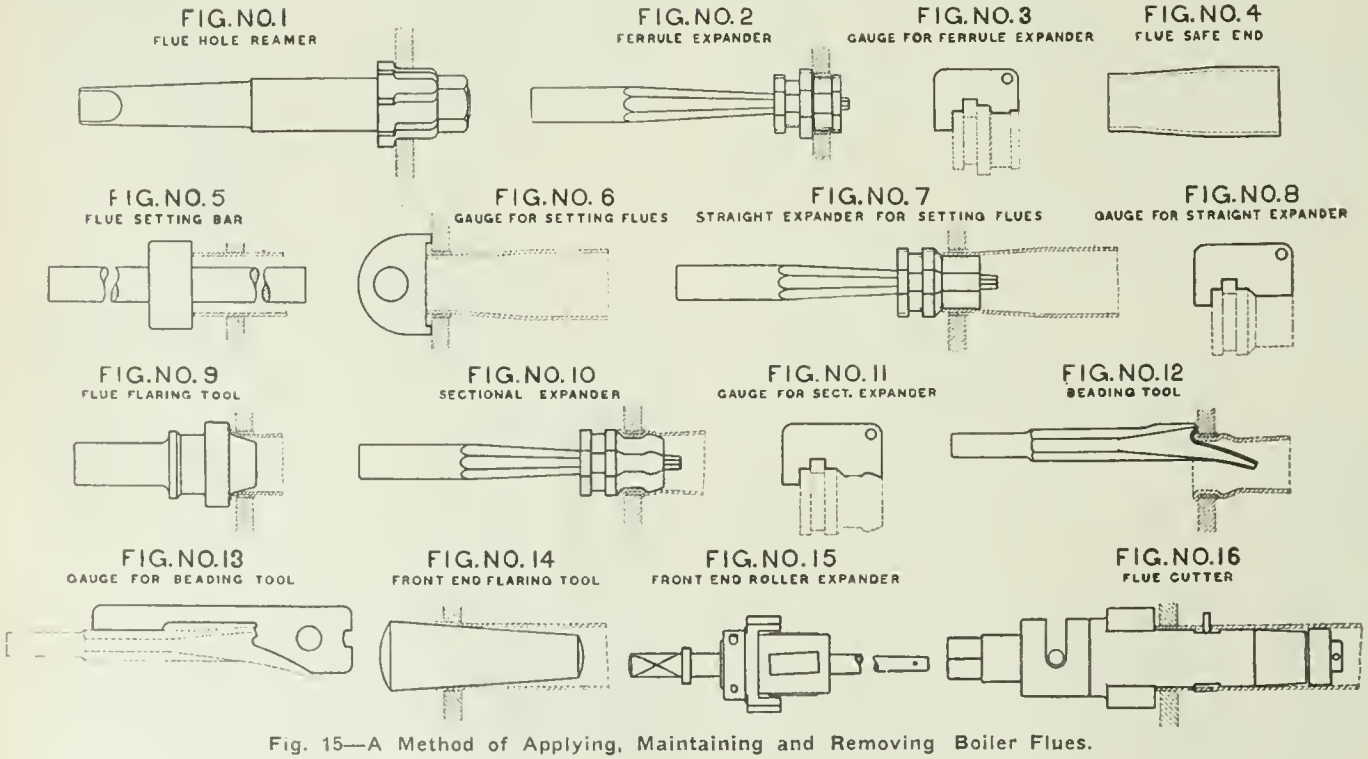


Fig. 15—A Method of Applying, Maintaining and Removing Boiler Flues.